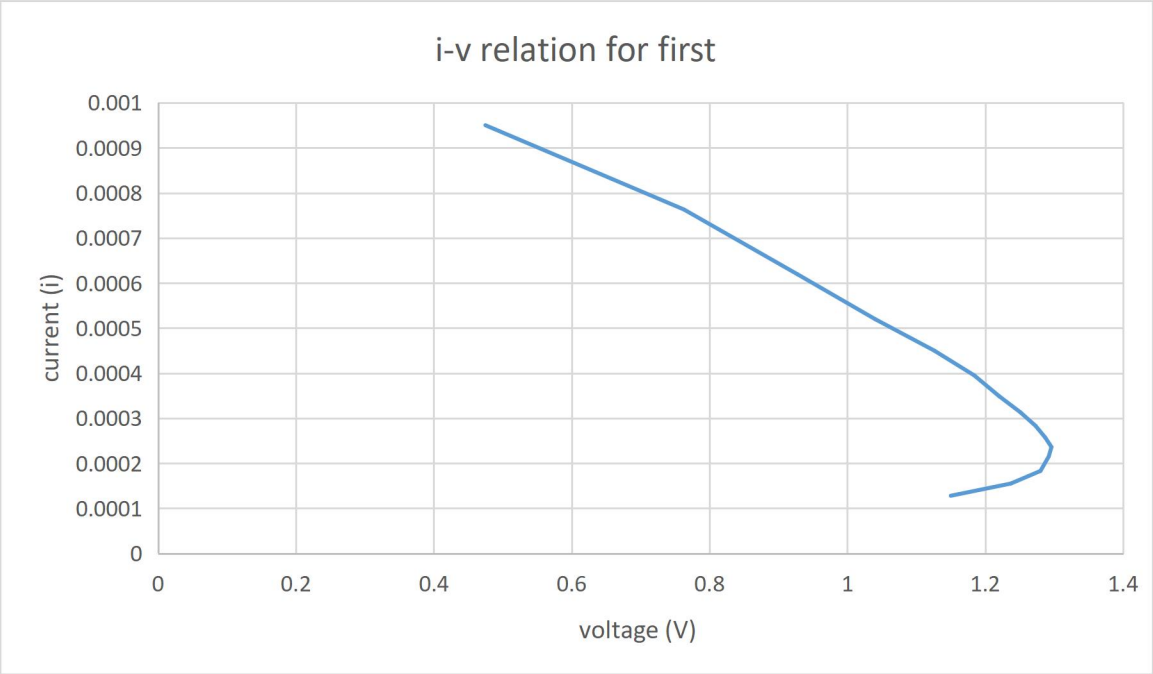
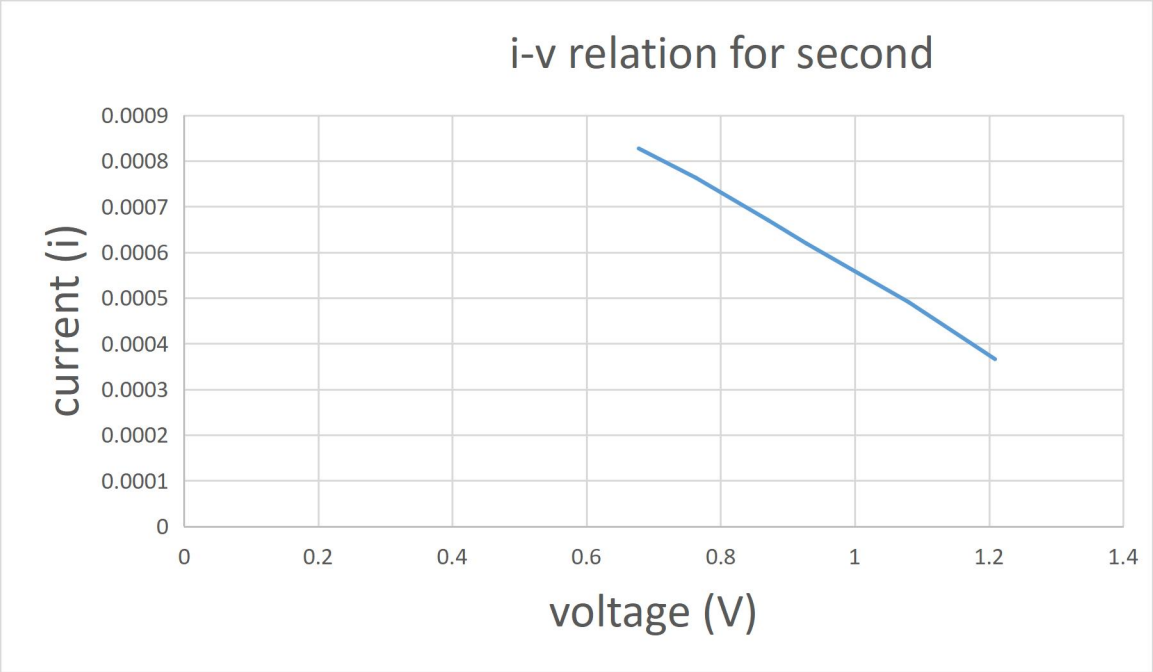
3.

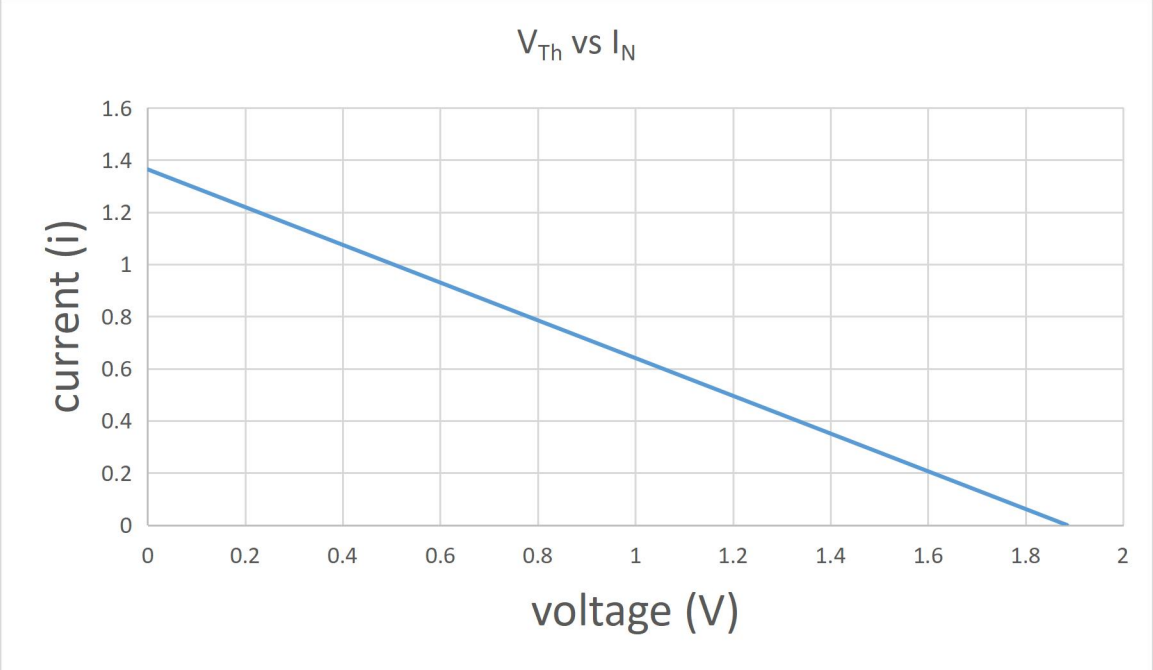
First			Second		
R	Vout	I	R	Vout	I
500	0.475	0.00095	820	0.678	0.000827
1000	0.763	0.000763	1000	0.763	0.000763
1500	0.928	0.000619	1300	0.871	0.00067
2000	1.04	0.00052	1500	0.928	0.000619
2500	1.126	0.00045	2200	1.08	0.000491
3000	1.184	0.000395	3300	1.209	0.000366
3500	1.22	0.000349			
4000	1.251	0.000313			
4500	1.273	0.000283			
5000	1.287	0.000257			
5500	1.296	0.000236			
6000	1.292	0.000215			
7000	1.28	0.000183			
8000	1.237	0.000155			
9000	1.15	0.000128			





$$V_{Th} = V_{open} = 1.885V$$

$$I_N = I_{sc} = 1.364mA$$

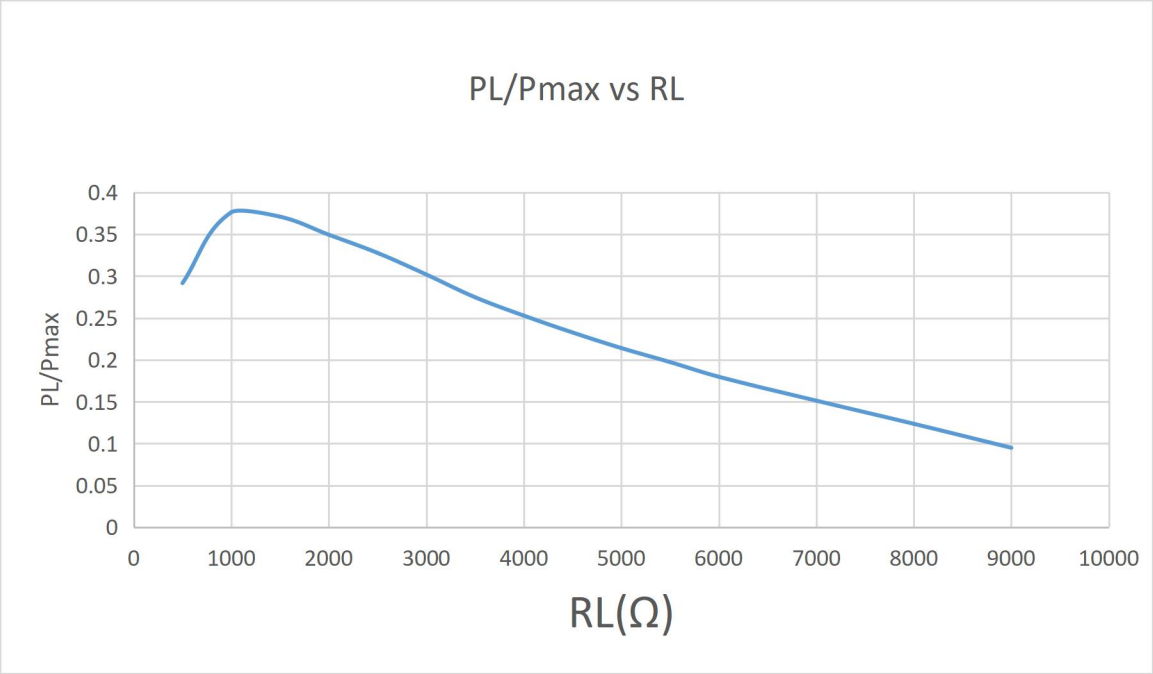


The i–v plots for both sections of the experiment display a linear relationship. However, in the first graph, once the voltage exceeds 1.296 V, it begins to decrease, which disrupts the expected trend.

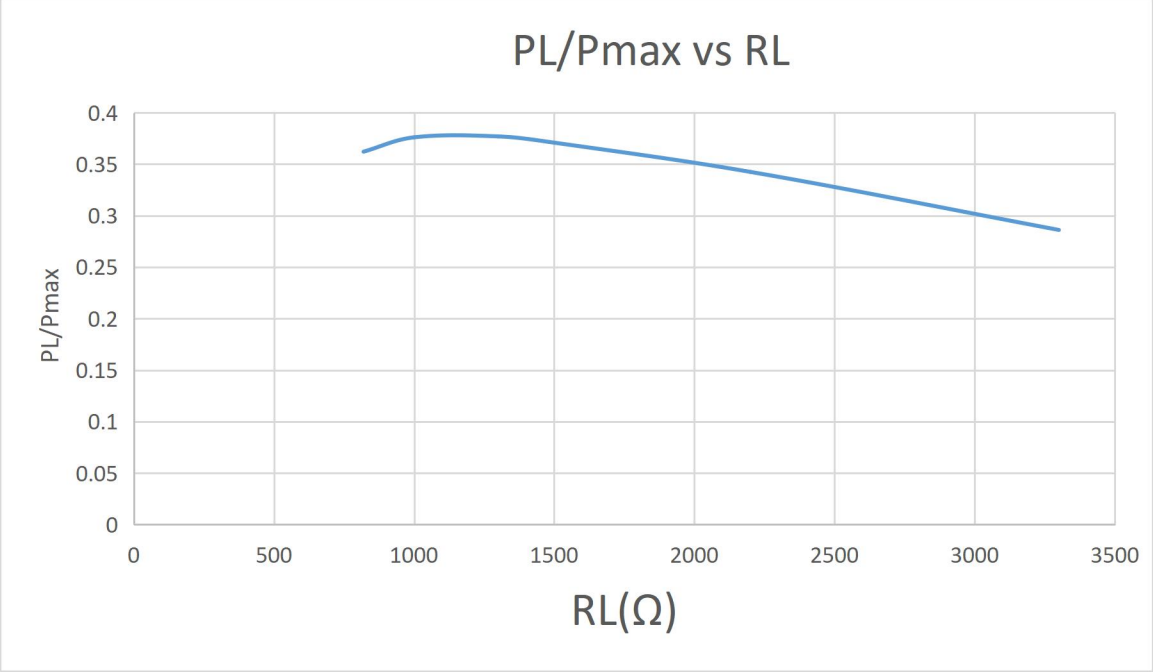
First Table						Second Table					
R	Vout	I	PL	PL/Pmax	Efficiency	R	Vout	I	PL	PL/Pmax	Efficiency
500	0.475	0.00095	0.000451	0.291505	0.095	ccc	0.678	0.000827	0.000561	0.362138	0.1356
1000	0.763	0.000763	0.000582	0.376078	0.1526	1000	0.763	0.000763	0.000582	0.376078	0.1526
1500	0.928	0.000619	0.000574	0.37088	0.1856	1300	0.871	0.00067	0.000584	0.376983	0.1742
2000	1.04	0.00052	0.000541	0.349354	0.208	1500	0.928	0.000619	0.000574	0.37088	0.1856
2500	1.126	0.00045	0.000507	0.327617	0.2252	2200	1.08	0.000491	0.00053	0.342495	0.216
3000	1.184	0.000395	0.000467	0.301864	0.2368	3300	1.209	0.000366	0.000443	0.286133	0.2418
3500	1.22	0.000349	0.000425	0.274714	0.244						
4000	1.251	0.000313	0.000391	0.252746	0.2502						
4500	1.273	0.000283	0.00036	0.232634	0.2546						
5000	1.287	0.000257	0.000331	0.214001	0.2574						
5500	1.296	0.000236	0.000305	0.197277	0.2592						
6000	1.292	0.000215	0.000278	0.179723	0.2584						
7000	1.28	0.000183	0.000234	0.1512	0.256						
8000	1.237	0.000155	0.000191	0.12356	0.2474						
9000	1.15	0.000128	0.000147	0.094925	0.23						

3.(b)

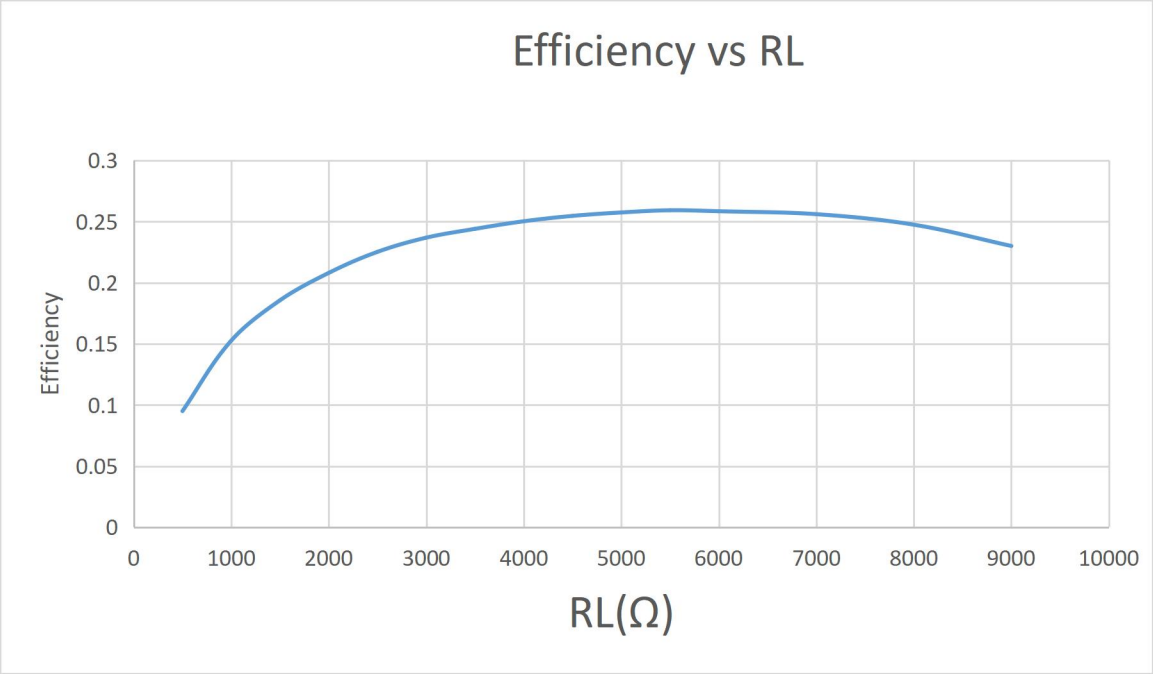
For first table



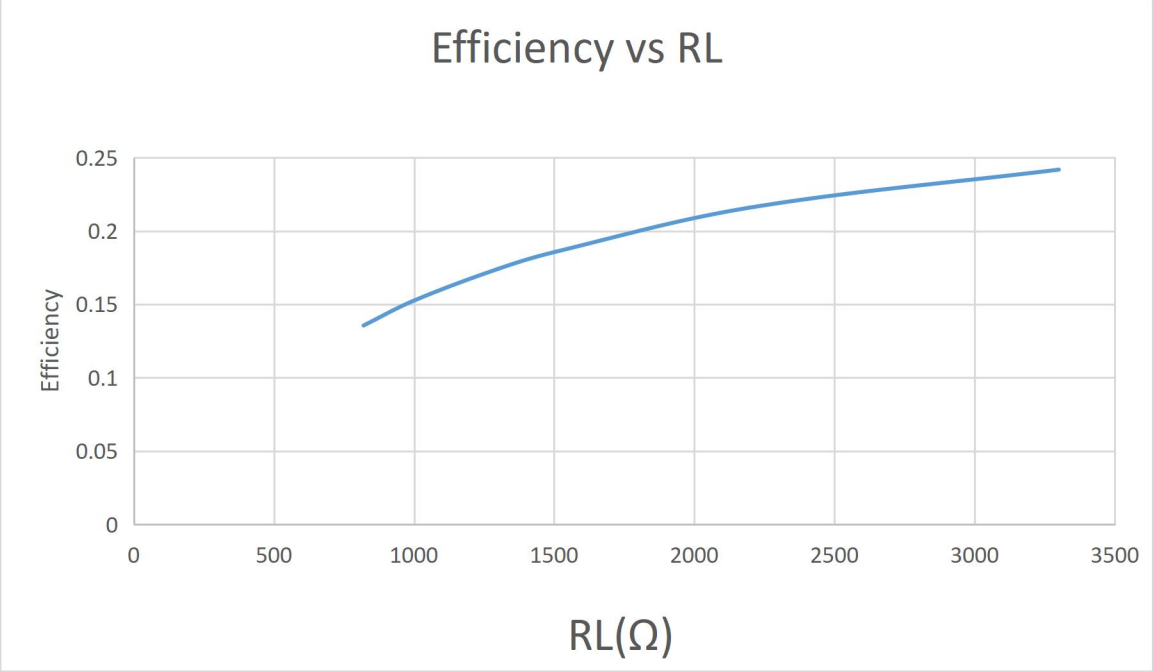
For second table

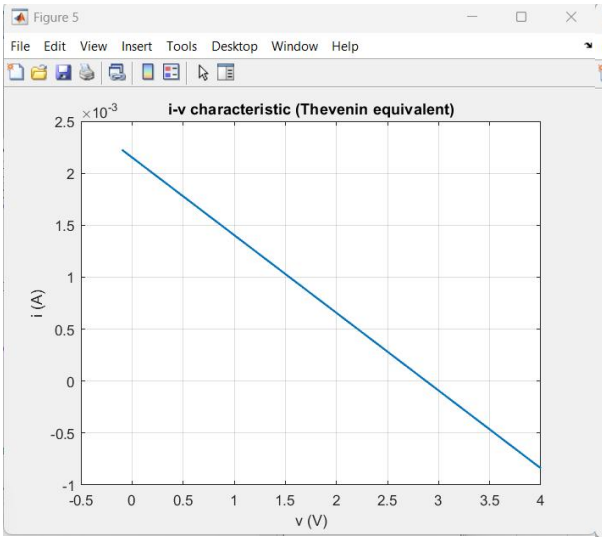


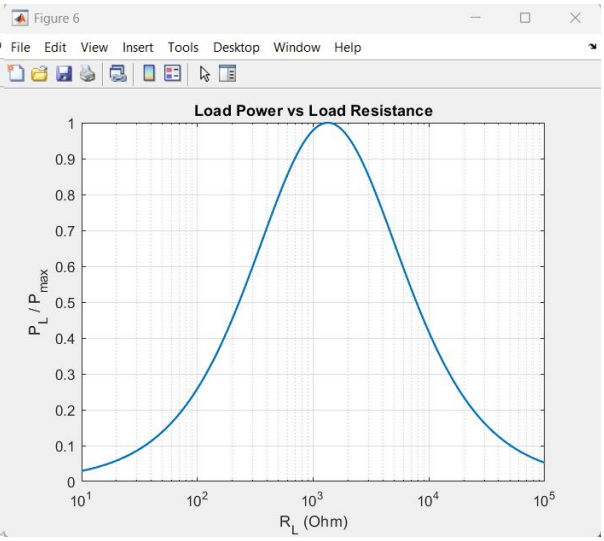
For First table

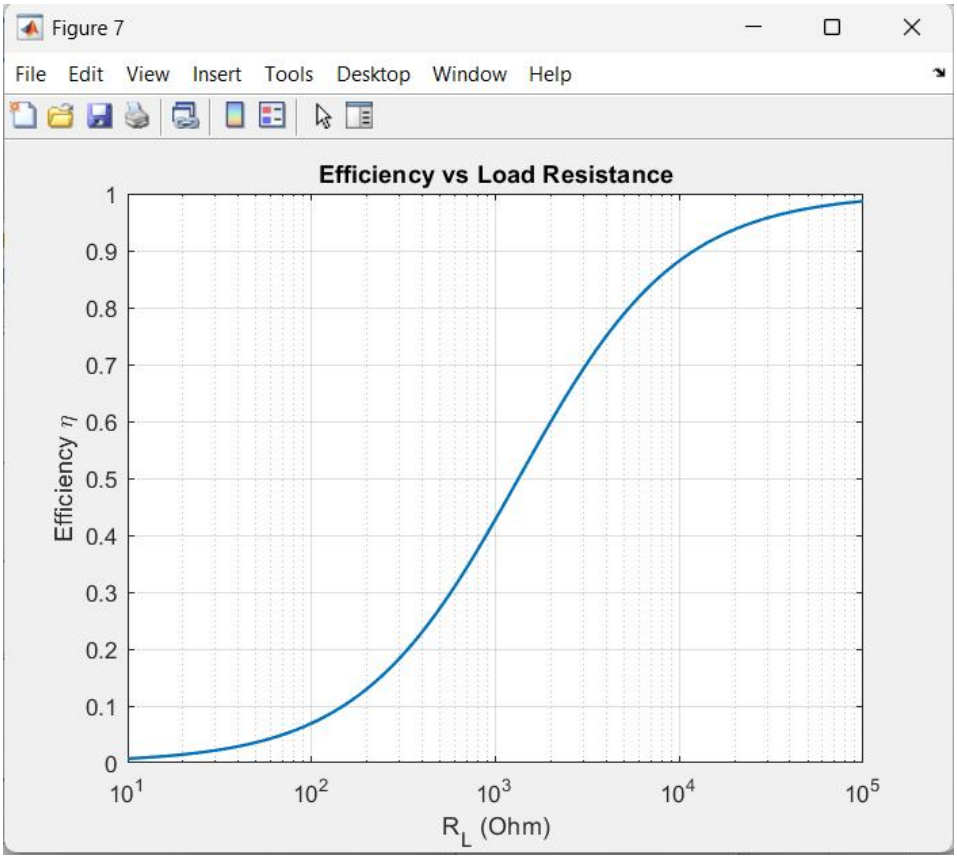


For second Table




Figure 5
File Edit View Insert Tools Desktop Window Help
i-v characteristic (Thevenin equivalent)
×10^-3
i (A)
v (V)


Figure 6
File Edit View Insert Tools Desktop Window Help
Load Power vs Load Resistance
P_L / P_max
R_L (Ohm)


Figure 7
File Edit View Insert Tools Desktop Window Help
Efficiency vs Load Resistance
Efficiency η
R_L (Ohm)

Thévenin / Norton equivalents & maximum power transfer

Open circuit voltage = 1.885V
Short circuit current = 1.364mA

Vin = 5V			
R	Vout	R	Vout
500	0.475	820	0.678
1000	0.763	1k	0.765
1500	0.928	1.3k	0.871
2000	1.040	1.5k	0.928
2500	1.126	2.2k	1.080
3000	1.184	3.3k	1.209
3500	1.220		
4000	1.251		
4500	1.273		
5000	1.287		
5500	1.296		
6000	1.292		
7000	1.280		
8000	1.237		
9000	1.150		

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